GCI

2026

HARDENING **NETWORK**

Small Cell CapEx Investment

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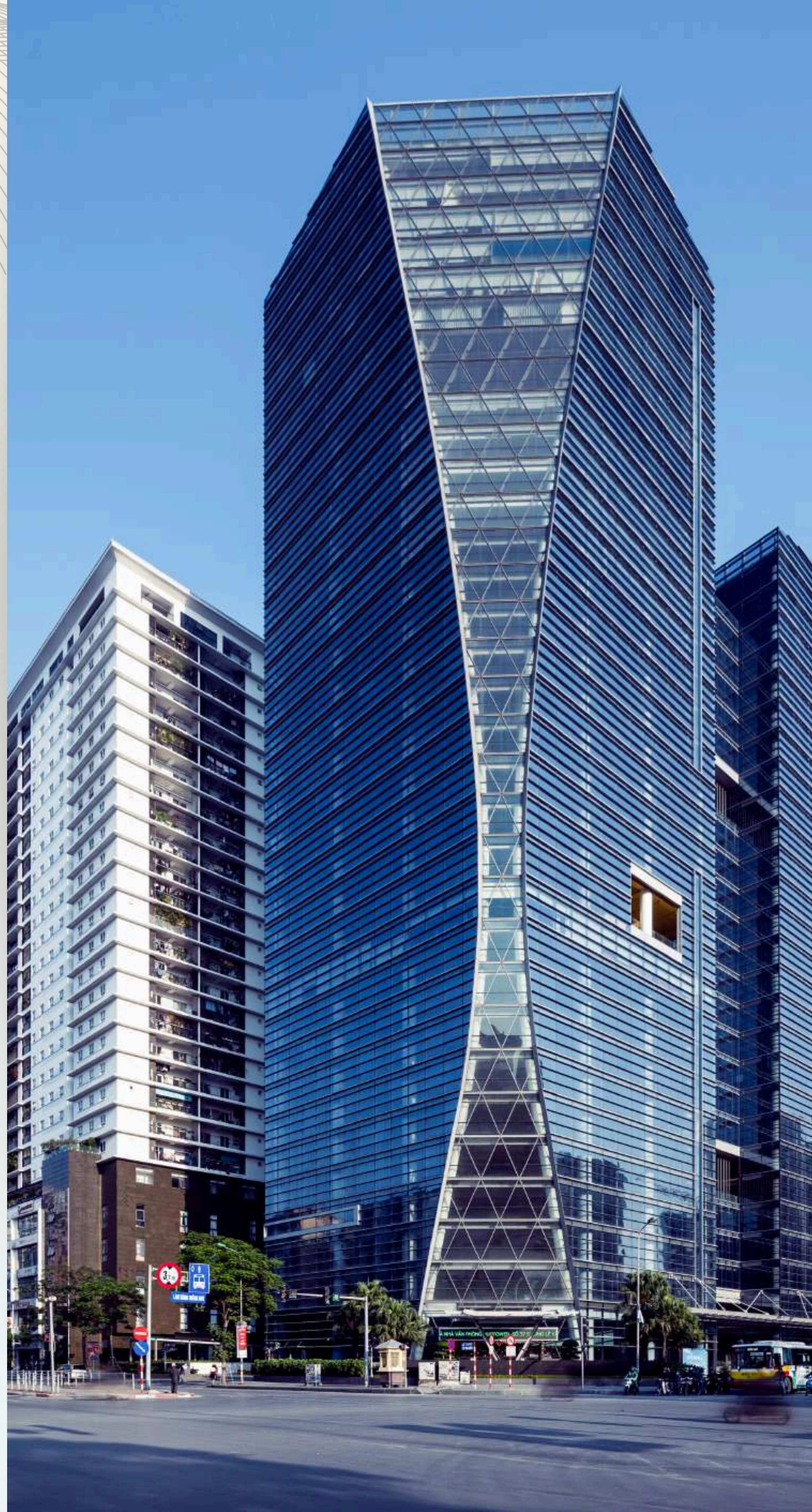


OPERATING MARKET CONTEXT

Market Value

Scaling from $17.14 Billion (2024) to $26.71 Billion by 2032 [1]

Key Churn Driver

Signal strength is the primary reason customers switch operators. [8]

Technical Proxy

To solve this churn threat, we analyzed a 100,000 record dataset from US operator. This data provides technical blueprint for exactly how network instability drives high value enterprise churn, and how we can prevent it.

While pricing drives acquisition, **network reliability** is the hygiene factor that dictates retention

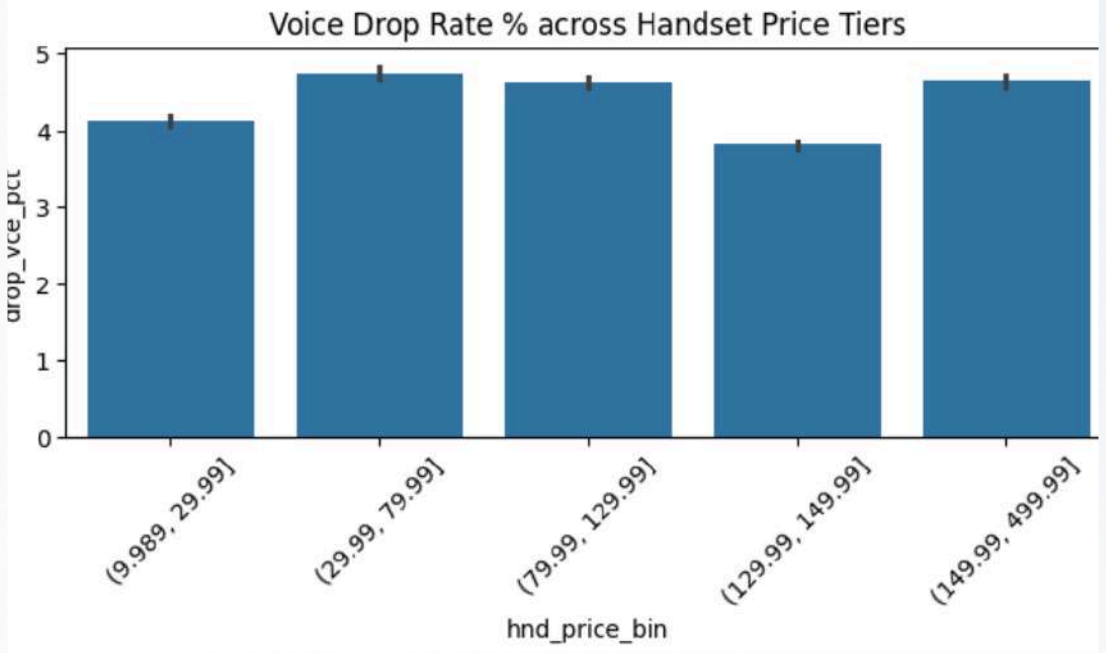
Technical Proxy: US carrier dataset

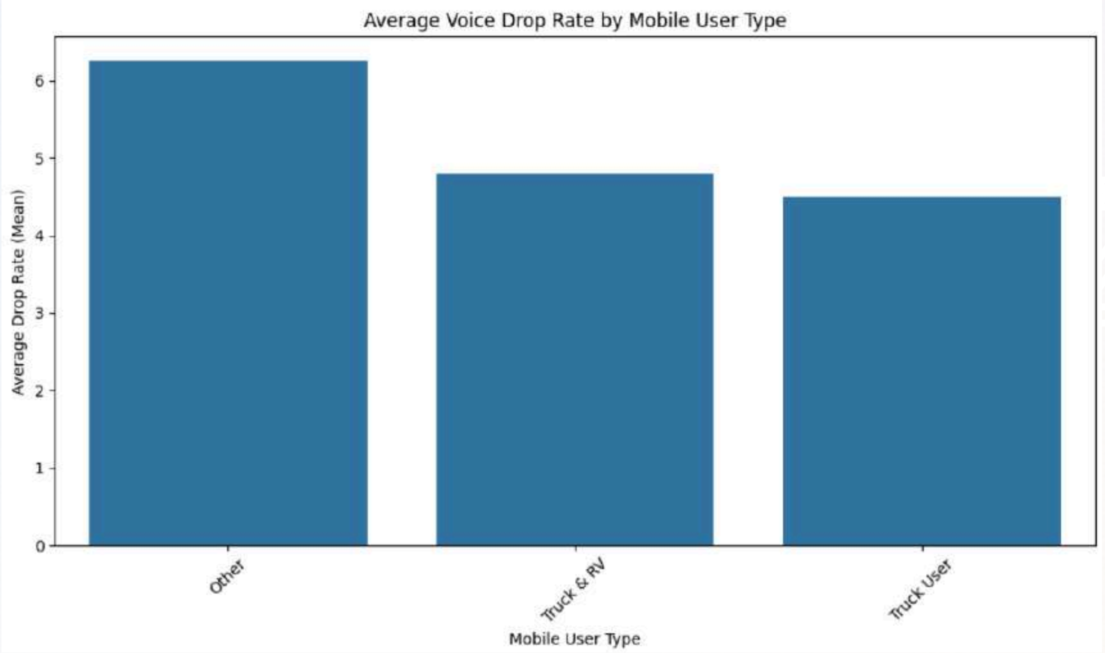
EXPLORATORY ANALYSIS

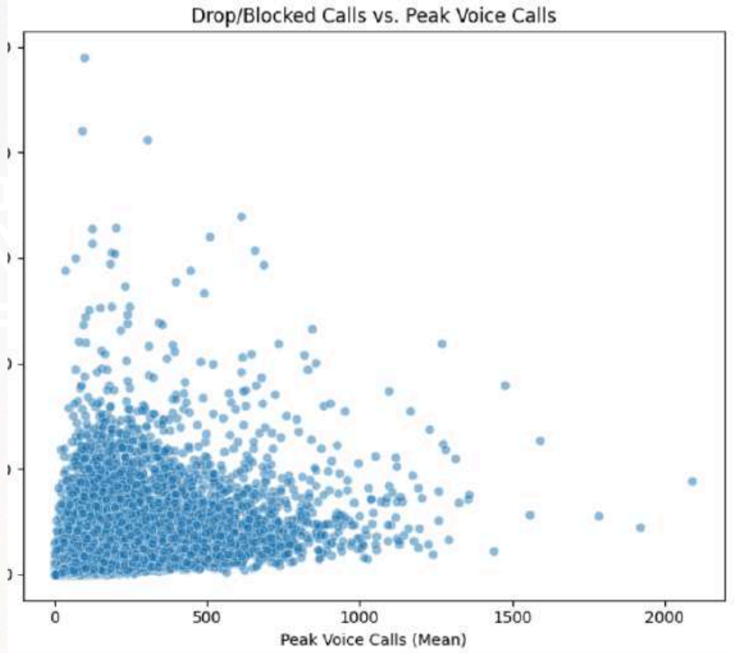
Dropped calls and blocked calls show significant variance and high averages, which are pain for any customer.

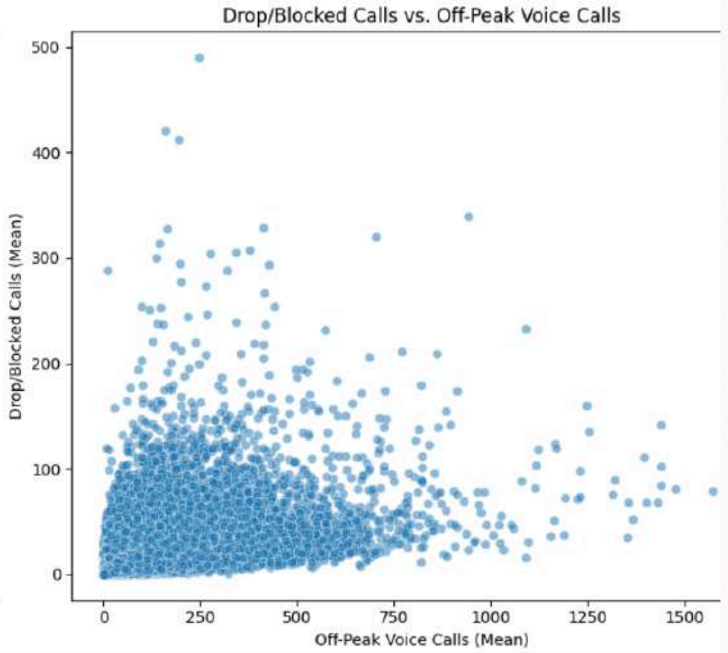
We will analyzing the target variable **drop_vce_Mean**

Suspected Cause	Data Evidence
Old/Cheap Handset Hardware	Voice drops occur equally across all handset prices and ages (new vs. refurbished)
Highway Mobility / Handoffs	RV/Truck users drop significantly fewer calls (avg 2.05) than stationary users (avg 5.46)
Peak-Hour Congestion	Mean drop rate is substantially higher during off-peak hours (0.263) than peak hours(0.148)





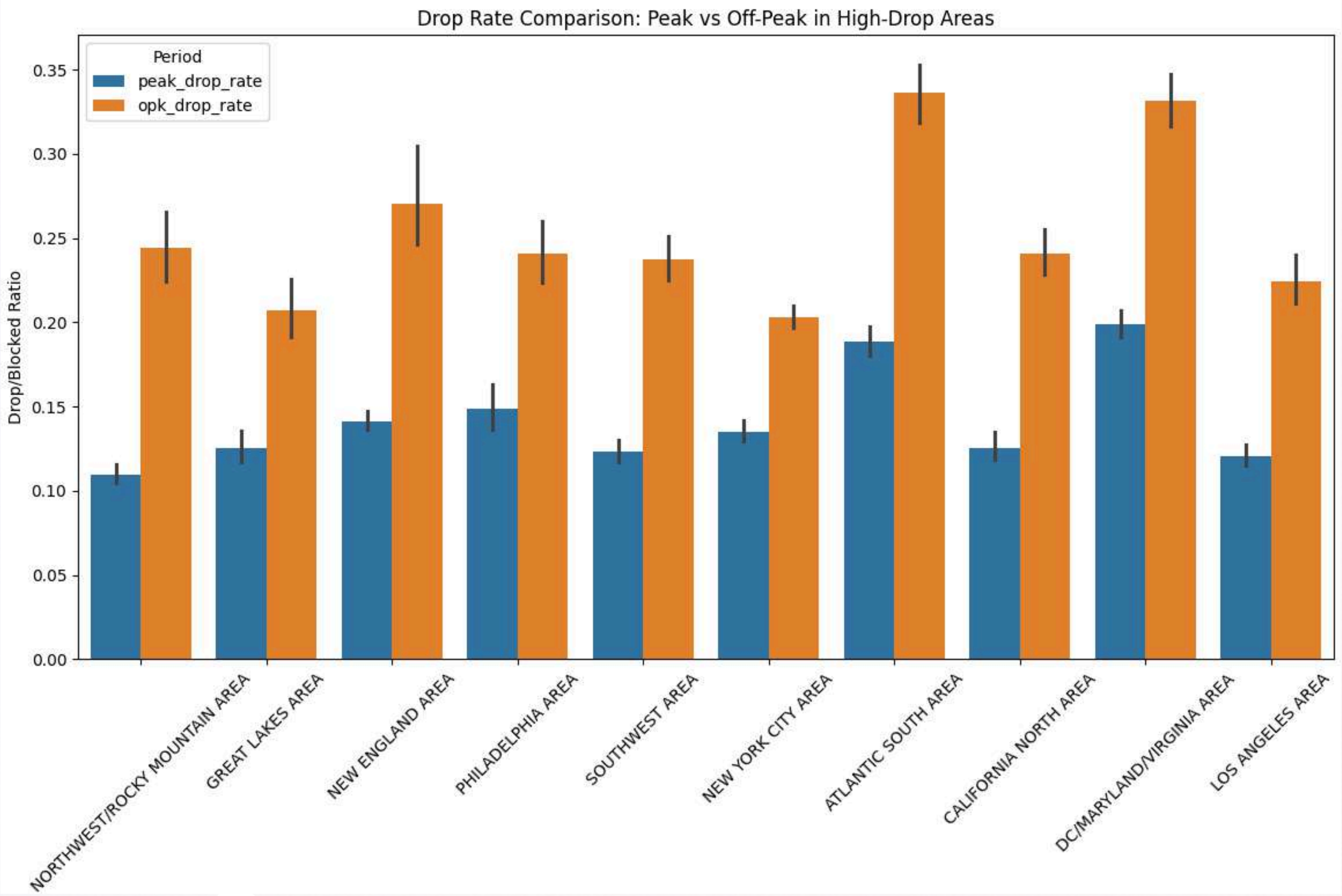




Technical Proxy: US carrier dataset

EXPLORATORY ANALYSIS

- if congestion from volume were the main driver, peak hours should be worse, not better.
- A specific group of busy cities like DC, Philly, and LA have much worse drop rates than both the biggest cities and smaller towns. This proves the problem isn't just "big city towers are too crowded" or "rural areas have bad signal.



With volume and demographic explanations weakened, the pattern that's left, off-peak-heavy, residentially concentrated, time-and-place specific, points toward signal penetration. Then it comes the hypothesis that

“The drop is caused by a **coverage** and **building-penetration issue** rather than a capacity issue”.

Technical Proxy: US carrier dataset

AI DIAGNOSTIC TOOL

Hence, 43% interpretability means the other 57% is explained by other variables or condition such as deeper layer of network, buildings, and much more. With error, ~4 calls.

Our AI tool isolates the operational drivers of network failure, separating user behavior from physical infrastructure gaps.

input

100,000+Customers record & 100 variables

AI Tool [5]

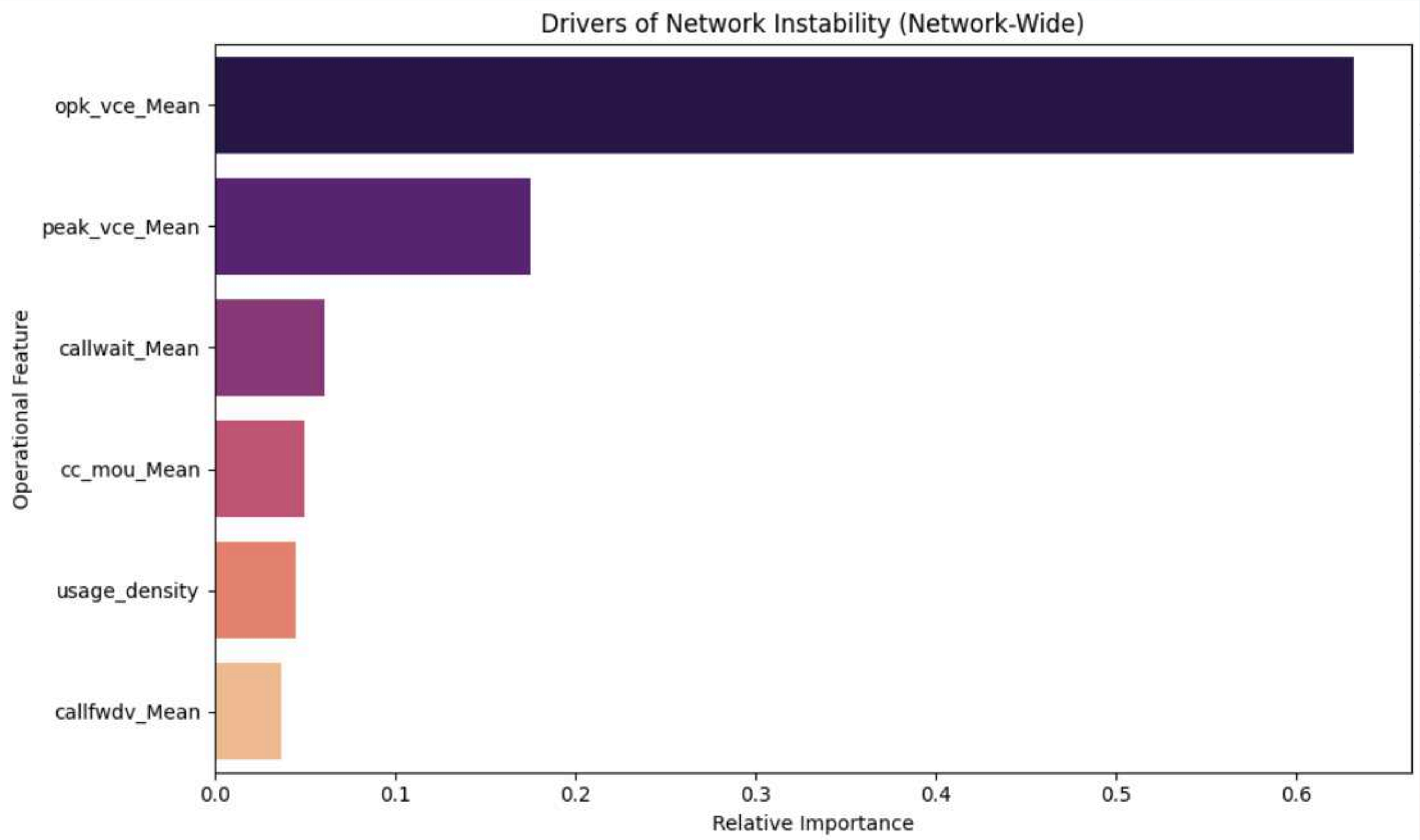
XGBoost Predictive Model. With accuracy score of r-squared: 0.43 (explains 43% of the variation in network drops) with 3.57 calls error margin. XGBoost was chosen for its robustness with mixed-type features and non-linear interactions

Output

Business drivers: Usage density and Signaling Stress

Technical Proxy: US carrier dataset

FEATURE IMPORTANCE



The AI explains 43% of drops through operational stress. The unexplained 57% isolates purely a physical problem “Infrastructure Dead Zones”

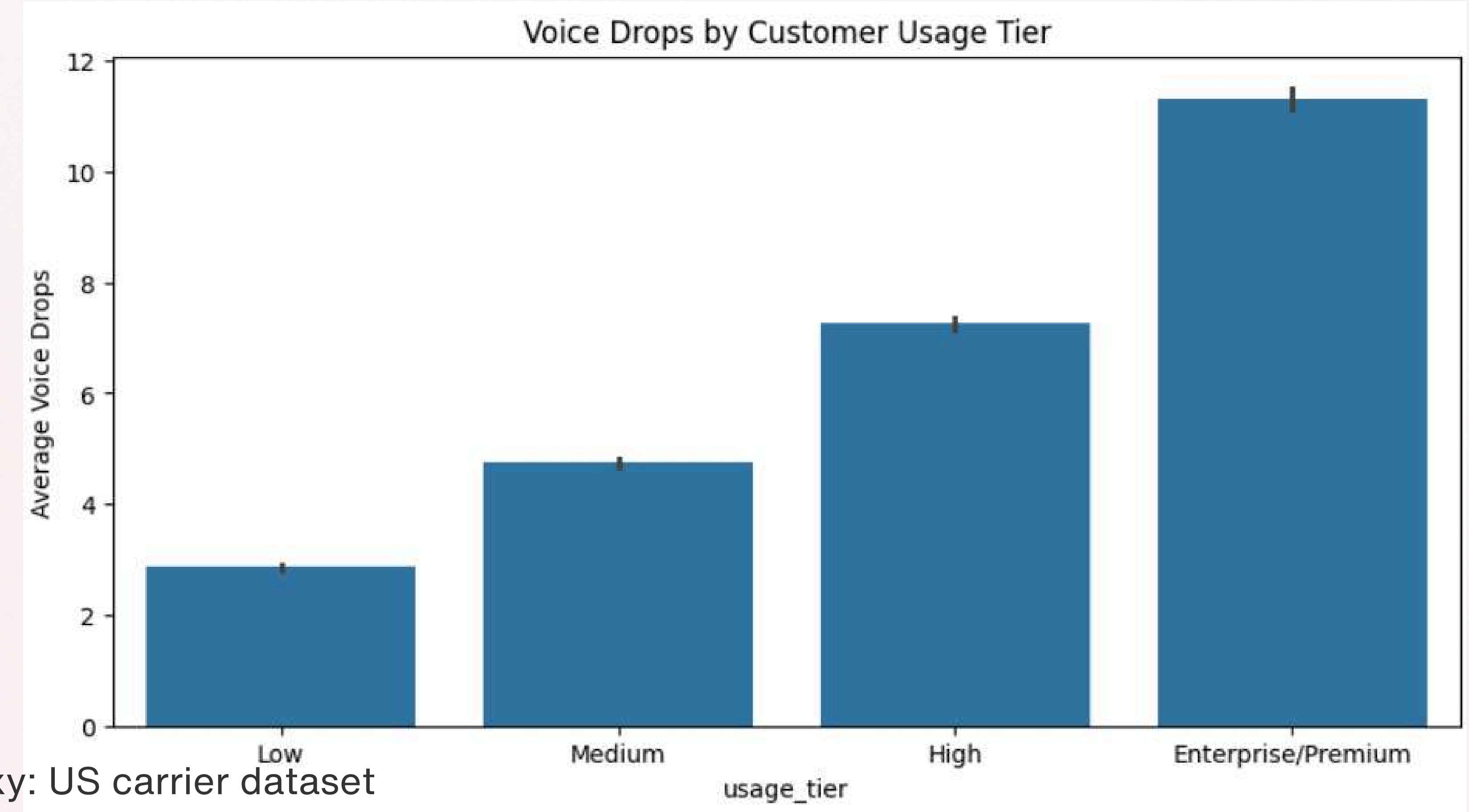
- *opk_vce_Mean* has roughly 0.66 relative importance, more than three times the weight of any other feature in the model. The **“Residential Dead Zone”** theory is now confirmed. Evening and weekend calling, when people are at home rather than in offices, is overwhelmingly more predictive of dropped calls than any other variable measured.
- Peak voice usage is the clear second driver at about 0.21. This confirms that daytime, business-hour traffic still stresses the network, but it's doing **roughly a third of the work** that off-peak usage is doing.
- Call waiting is top 3 major predictor, the network architecture struggles to hand off concurrent calls. While it’s not significant
-

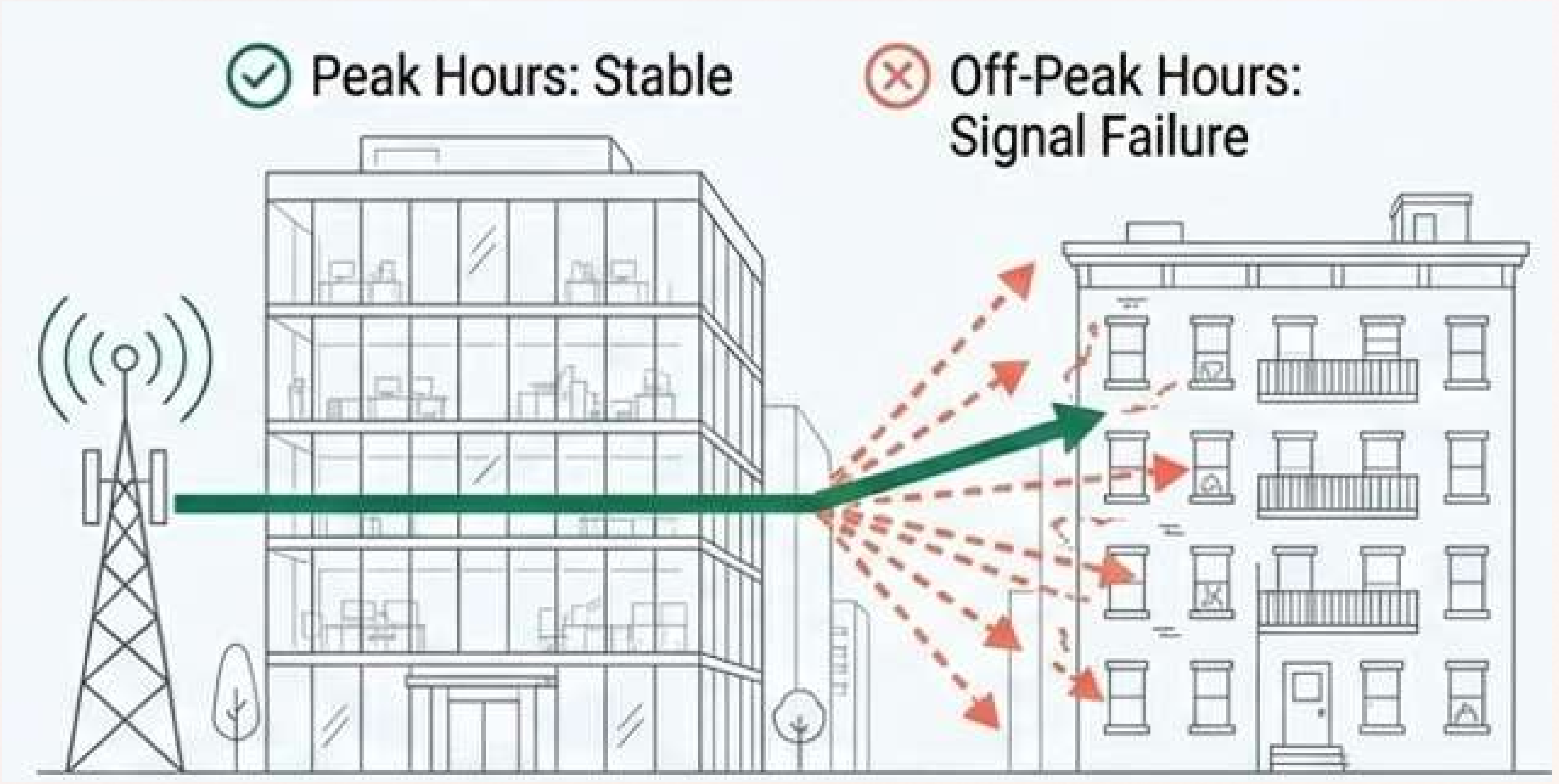
Hence **Wi-Fi calling adoption** and **indoor coverage** need to be the top engineering and marketing priority, not one priority among several.

Technical Proxy: US carrier dataset

PHYSICAL COVERAGE GAP

- The fragile areas (NYC, Philadelphia, LA) have high income, dense population, and terrible network efficiency.
- Our network isn't failing because it's too busy, it's failing because it can't handle complex signaling for our highest value enterprise clients before they go inside their urban or homes.
- This is physical penetration and coverage gap, not a temporary capacity issues.





Technical Proxy: US carrier dataset

FINANCIAL EXPOSURE

Target Segment

Enterprise / Premium Tier
Experience the highest instability (11.3 average drops vs 2.8 for low tier) and generates the highest margin

The Geography

The Fragile Zones
New York, Philadelphia, LA, New England.

Current churn in these zones already elevated (51.3% vs 48.8%)

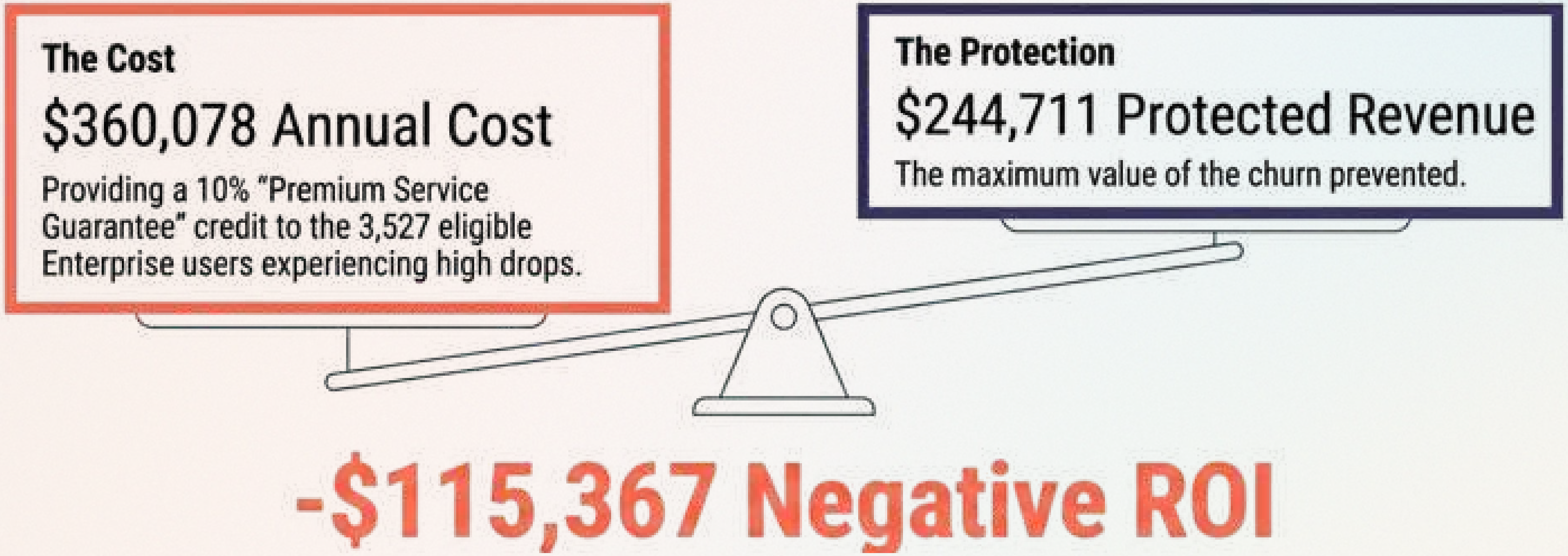
Value at Risk

Value at Risk $407,851 MRR, 5% churn increase → $244,711 annual loss.

Conservative 5% increase in Enterprise churn due to this network fragility equals $244,711 in lost annual revenue

Technical Proxy: US carrier dataset

CREDIT TRAP



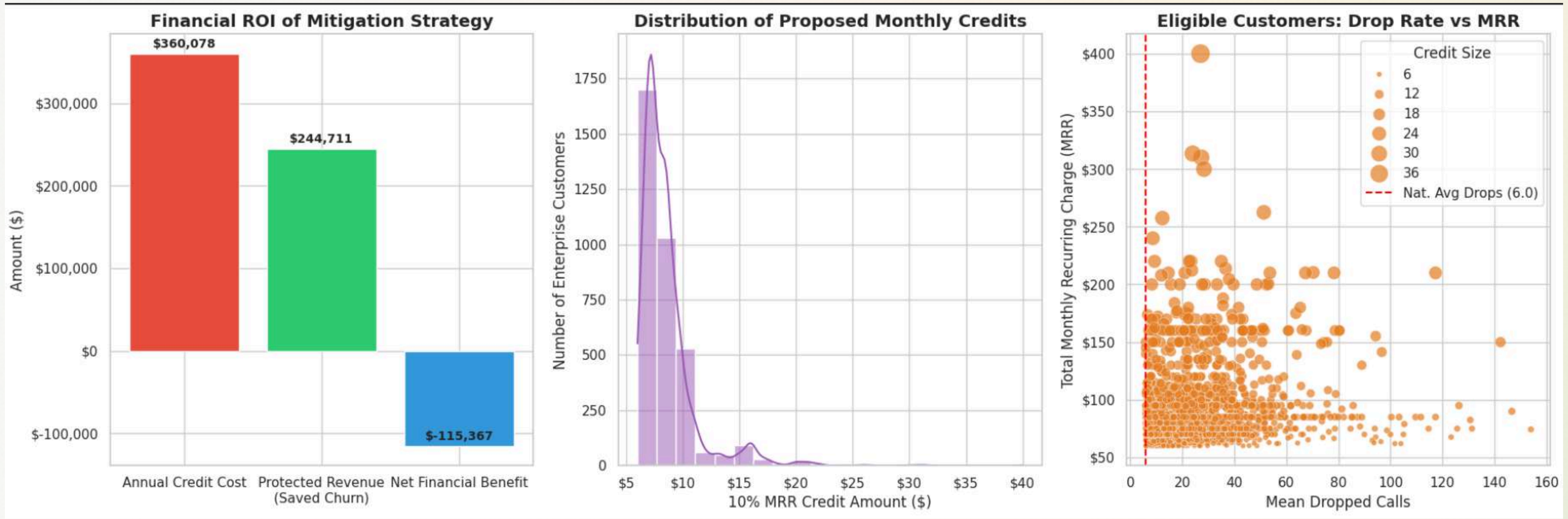
Trying to buy some “quick” path isn’t the best option. Proactive credit is far more expensive than the churn we prevent.

Technical Proxy: US carrier dataset

DEPLOYMENT PRIORITY & ROOT CAUSE

The priority is shown above, with the root cause shown

	area	primary_states	priority_score	total_mrr	likely_root_cause
0	NEW YORK CITY AREA	NY, NJ, CT	1.000000	508289.551667	Infrastructure/Capacity
1	LOS ANGELES AREA	CA	0.758767	305939.852500	Infrastructure/Capacity
2	NEW ENGLAND AREA	MA, CT, RI, NH, ME, VT	0.682333	240162.331667	Infrastructure/Capacity
3	PHILADELPHIA AREA	PA, NJ, DE	0.672768	110289.978333	Software/Signaling
4	NORTHWEST/ROCKY MOUNTAIN AREA	NaN	0.560810	194967.135833	Infrastructure/Capacity



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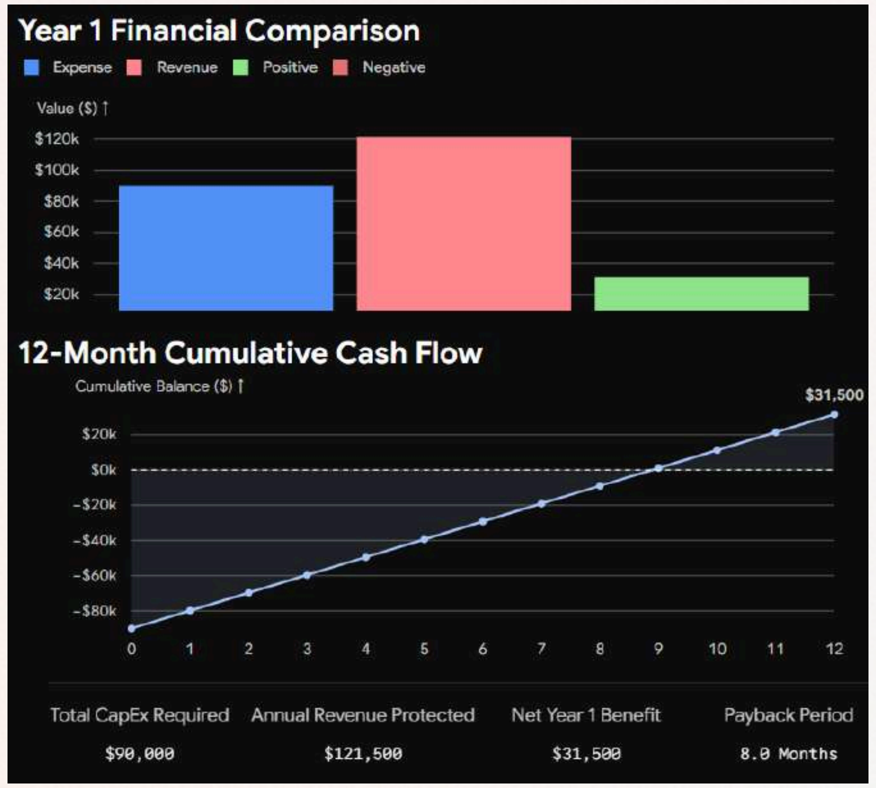
ROADMAP

Phase 1

Initiate RF (Radio Frequency) building penetration testing specifically in the New York City and Los Angeles target quadrants.



Cost estimation

At an estimated fully-loaded cost of $12,000 per unit, this secures budget for 30 indoor Small-Cell nodes to be deployed in our most critical high-value enterprise buildings.

We may have to ask permit to local government, while waiting for a time, we should use the existing mobile app to push notification to turn on WiFi Calling

Phase 2 Deployment

Deploy with the budget, and most vital part of the state, and test using post review & forms

Phase 3 Evaluation

- Coverage Gap: The lack of variance between peak and off-peak drop rates in these areas confirms that the issue is physical 'dead zones', not temporary network congestion.
- Signaling Stress: Deployment should include signaling software optimization to handle the high volume of Call Waiting/Forwarding typical of these high-value professional segments.

Technical Proxy: US carrier dataset

PROPOSAL

What

We are proposing a targeted **Small-Cell Deployment Roadmap.** Instead of issuing reactive billing credits or "Premium Service Guarantees" for dropped calls, we are going to redirect that budget into **permanent**, physical infrastructure hardening

Where

We are prioritizing Capital Expenditure (CapEx) strictly in our top "Fragile Zones", the markets that represent the highest intersection of network failure and revenue volume.

Why

Upgrading infrastructure is the only financially viable path to protect this revenue, it succeeds eliminates indoor dead zones. Potentially protects $244K annually. Also it is permanent.

When

Immediately. Our risk assessment shows that churn in these fragile zones (51.3%) is already outpacing our stable zones (48.8%). This indicates that our Enterprise customers' "tolerance phase" is actively ending right now. We must initiate deployment before these clients reach their contract renewal periods.

Technical Proxy: US carrier dataset

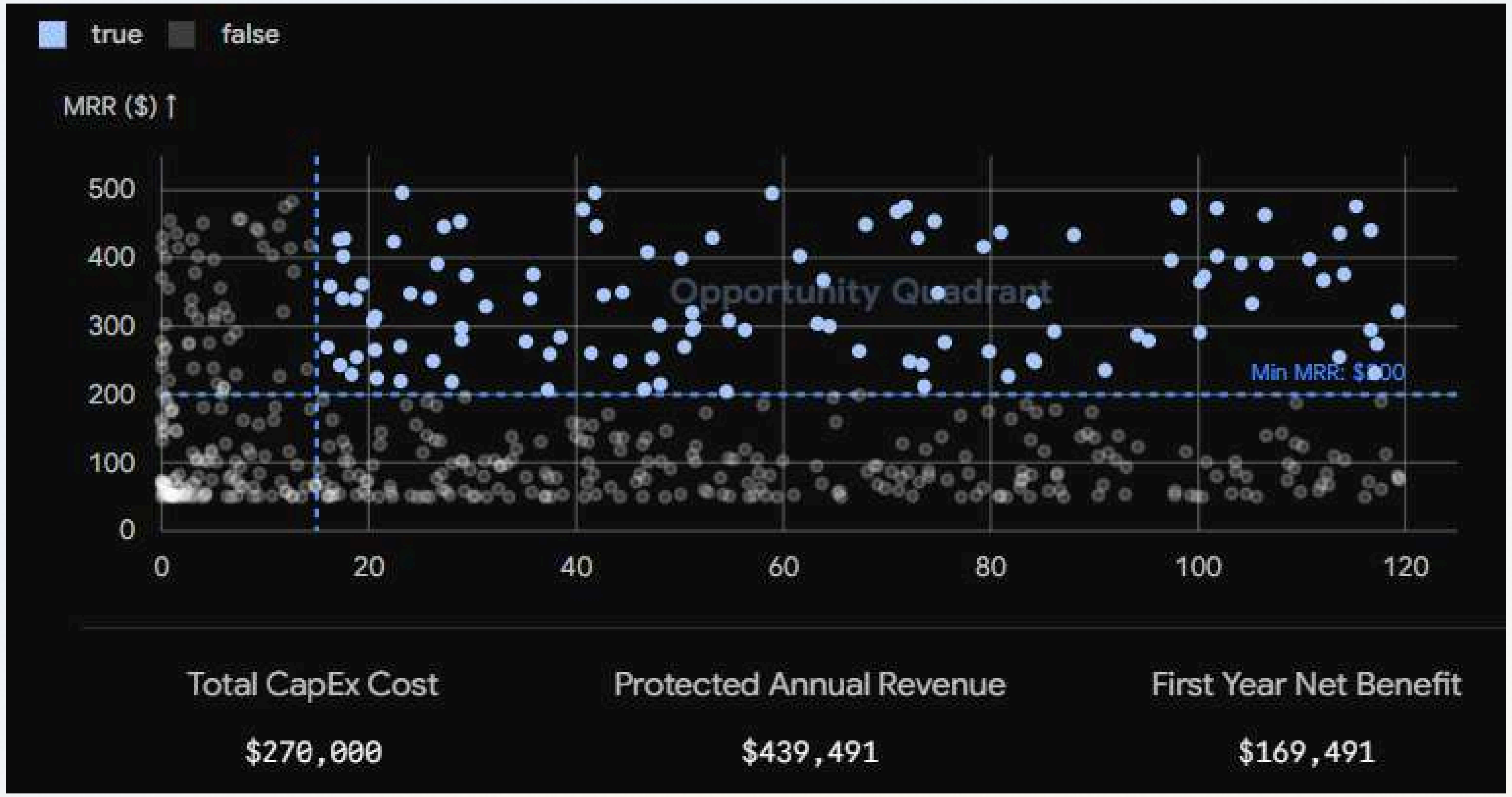
BENEFITS AND IMPACT

Financial Impact

- Rapid payback Period
- One time cost - permanent save
- Asset Creation

Operational & Strategic Impact

- Root cause resolution, dropping customer churn
- Stimulate new contract renewal



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